Cogtics – Full Technical Appendix

Production-Ready MVP: Architecture, Data Model, Rules, and Operations

A1. System Architecture

Cogtics is built as a modular monolith with clear service boundaries. Core layers include Ingestion Connectors, Canonical Normalization, Contract Intelligence, Revenue Recognition, Detection Engine, Simulation Engine, Workflow & Evidence, and Export Services. All write operations are internal only; external systems are read-only.

A2. Connector Contracts

Each connector implements a standard contract: authenticate, discover schema, incremental sync, error reporting, rate limit handling, and replay. Connectors are versioned independently.

A3. Canonical Schema Overview

Core tables include tenants, connections, sync_runs, snapshots, customers, contracts, contract_versions, contract_terms, invoices, invoice_lines, payments, revenue_obligations, revenue_schedules, recognition_events, issues, issue_evidence, and audit_logs.

A4. Identity Resolution Logic

CRM is the primary source of customer identity. Billing and accounting entities are matched using deterministic rules with assisted human confirmation for conflicts.

A5. Contract Parsing & Storage

Contracts are stored in object storage and parsed into clause-level terms. Each term is stored with confidence, source offsets, and extracted values. Overrides create new contract versions.

A6. Revenue Recognition Engine

The engine supports performance obligation identification, transaction price allocation, straight-line and usage-based recognition, deferred revenue tracking, and reconciliation against invoices and GL entries.

A7. Detection Rules Framework

Detection rules are deterministic, pure functions operating on snapshots. Each rule emits issues with rule_id, version, expected vs actual values, severity, confidence, and evidence.

A8. Guaranteed Leak Rules (MVP)

Missing invoice, underbilling, overbilling, expired discount applied, billing beyond contract term, duplicate billing, wrong price, wrong period, currency mismatch, recognized vs invoiced mismatch.

A9. Simulation Engine

Simulation mode clones a snapshot and applies parameter overrides. No source data is mutated. Outputs include delta revenue, timing shifts, and recognition impact.

A10. Evidence Model

Each issue links to source records, calculations, contract clauses, and timestamps. Evidence is immutable and audit-grade.

A11. Workflow State Machine

Issue lifecycle: open, resolved, dismissed. All transitions are logged. Role-based access controls restrict actions by user role.

A12. Export Architecture

CSV exports are generated directly from snapshot queries. PDFs are rendered server-side. Email digests are scheduled jobs with configurable frequency and scope.

A13. Security Architecture

Tenant isolation is enforced at the database layer. Encryption in transit and at rest is mandatory. Audit logs capture all access and mutations. Secrets are stored in managed key services.

A14. Performance & Scaling

Async jobs handle ingestion and detection. Indexing and partitioning support large datasets. Backfills are chunked and resumable.

A15. Observability & Ops

Metrics, structured logs, traces, and alerts are enabled. Feature flags support safe rollouts and rollback. Connectors are monitored independently.

A16. Failure & Recovery

Failures are isolated per tenant and per connector. Idempotency keys and checkpoints allow safe retries. Snapshot replay supports audits and debugging.

A17. Non-Goals (Explicit)

No write-back to source systems. No billing execution. No tax calculation. No full ERP replacement.